

Jalen Edusei

jalen.edusei@gmail.com | Atlanta, GA | jalenedusei.com | linkedin.com/in/jalenedusei | github.com/jke48222

Full-stack software engineer shipping production web applications for paying clients: React/TypeScript frontends, REST and realtime services on PostgreSQL, Docker/Kubernetes deployment, and AI features built directly on the Claude API.

EDUCATION

University of Georgia, Morehead Honors College, Athens, GA
Bachelor of Science in Computer Systems Engineering, cum laude

May 2026
GPA: 3.61 / 4.00

Relevant Coursework: Data Structures, Systems Programming, Software Development, Discrete Mathematics, Embedded Systems I & II, Advanced Digital Design, Human-Computer Interaction

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Swift, Elixir, Java, C, C++, SQL

Frontend: React 19, Next.js (App Router, ISR), Tailwind CSS, three.js / React Three Fiber, Framer Motion, WCAG AA accessibility

Backend & Data: Node.js, Phoenix / Ecto, PostgreSQL (row-level security, transactional outbox), Supabase, REST / JSON APIs, WebSockets, SQLite / FTS5, Docker, Kubernetes, GitHub Actions, Vercel

AI & Practices: AI-paired development (Claude Code, Cursor), Anthropic Claude API (tool use, streaming, structured outputs), MCP servers, Agile team development, code review, automated testing (Vitest, ExUnit, XCTest)

EXPERIENCE

Freelance Software Engineer, *Self-Employed*, Atlanta, GA

May 2026 – Present

- Designed, built, and shipped kulenterprises.com for a Georgia freight carrier: 22 statically generated Next.js 16 / React 19 / TypeScript pages on a git-backed TinaCMS content model, a dependency-free CMS-scoring site search, four rate-limited spam-protected forms through Resend, and WCAG AA-audited contrast.
- Extended the engagement with a broker credit-vetting portal on Supabase: a 12-question screening workflow, risk ratings with an admin-only do-not-load block, credit-exposure checks with recorded approvals, and compliance rules held in Postgres (row-level security, audit log, atomic SQL functions) behind invite-only auth with four staff roles.
- Built and shipped akilahmali.com for a recording artist: a real-time 3D control-room scene in three.js / React Three Fiber, a Web Audio rotary payphone, and a roughly 46 MB payload cut through code-splitting, with strict security headers and an email-capture API route.

Capital One, *Business Analyst Intern*, McLean, VA

Jun 2025 – Aug 2025

- Built the business case for a CreditWise Notifications Preferences Center projected to centralize communication management for 60M+ customer accounts; presented the recommendation to senior leadership.
- Analyzed email-campaign performance in SQL, Python, and Excel, creating a valuation framework that quantified engagement and retention impact and informed product-roadmap decisions.

UGA Small Satellite Research Lab, *Flight Software Developer (MEMESat-1)*, Athens, GA

Mar 2024 – Dec 2024

- Developed and debugged embedded C/C++ flight software on STM32 microcontrollers for the MEMESat-1 CubeSat, bringing up UART communication between subsystems and hardening it with unit and integration tests.

PROJECTS

Relay: Order Management & Fulfillment | *Elixir, Phoenix, PostgreSQL, React, TypeScript, CI/CD*

Jul 2026

- Built an event-driven order management system: 13-route JSON API, Kafka-style transactional outbox, 8-state machine enforced under row locks, crash-safe worker with boot-time rehydration, and a live React dashboard over Phoenix Channels.
- Prevented oversell with SELECT ...FOR UPDATE allocation across four facilities plus a CHECK constraint and Idempotency-Key replay; 52 failure-mode tests (ExUnit, Vitest) run in GitHub Actions; Docker and Kubernetes manifests.

Live Election Platform | *Next.js 14, JavaScript, Supabase (Postgres, Realtime, RLS)*

Mar 2026 – Jun 2026

- Shipped a realtime chapter election system (14 roles, 39 candidates): three-state election machine, SHA-256 device fingerprinting with a UNIQUE(role_id, device_hash) constraint and row-level security, and a 29-finding security audit verified by 42 HTTP-level tests.

Exocortex: Local-First Personal Memory System | *Swift, SQLite/FTS5, MLX, Foundation Models, MCP*

Aug 2026

- Built Exocortex, a local-first memory system that gives AI tools one private, shared memory: 14 life-log streams flow into a 100,000-event SQLite/FTS5 store with deduplication, secret redaction, trust-class retention, and encrypted, restore-verified backups.
- Engineered hybrid search fusing BM25 with a binary vector index over on-device embeddings, measuring 0.95 Recall@1 against a 0.55 keyword-only baseline, and exposed it through an MCP server with per-client trust tiers and sanitization against data exfiltration.

Otto: AI Assistant in the MacBook Notch | *Swift, SwiftUI, AppKit, Anthropic API*

Sep 2026

- Built a macOS assistant in the MacBook notch that streams Claude API replies over SSE with web search and citations, modeling hover and click-through as a pure state machine (24 tests); 16,400 lines of dependency-free Swift, 212 unit tests.

LEADERSHIP & HONORS

Vice President, NSBE UGA Chapter (100+ members) | Tau Beta Pi | Extraordinary Engineer Award | 2x Presidential Scholar | Dean's List